# American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Junior Division

**1. Prefix/Infix/Postfix Notation**

Evaluate the following prefix expression if all numbers are single digits:

+ − * 2 3 / 4 2 ^ 2 3

A. 14
B. 4
C. 12
D. 10
E. 6

**2. Prefix/Infix/Postfix Notation**

Convert the following infix expression to postfix:

$$\frac{A+B}{C} - \frac{B*C}{A-C}$$

A. A B + C / B C * A C − / −
B. A + B C / B C * A C − / −
C. A B C + / B C * A C − / −
D. A B + C / B * C A C − / −
E. A B + C / B C * A − C / −

**3. Bit-String Flicking**

Evaluate:

10100 OR 01100 AND 01011

A. 01000
B. 11100
C. 00111
D. 11111
E. 00000

**4. Bit-String Flicking**

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 01110)))

A. 00010
B. 10001
C. 00001
D. 00100
E. 01100

**5. What Does This Program Do? (Looping)**

What is the following program's output for the given input values?
p = 1000, r = 10, t = 3

```
input p, r, t
o = p
for c = 1 to t
     i = p * r / 100
     p = p + i
next c
f = p - o
output f
```

A. 100
B. 110
C. 121
D. 331
E. 1331

# American Computer Science League

2022-2023 ● Contest 2: Short Problems ● Junior Division

**1. 前缀/中缀/后缀**

如果下方所有数字都是一个单独的数值，请计算下方前缀表达式：

+ − * 2 3 / 4 2 ^ 2 3

A. 14
B. 4
C. 12
D. 10
E. 6

**2. 前缀/中缀/后缀**

请将下方中缀表达式转换为后缀表达式：

$$\frac{A+B}{C} - \frac{B*C}{A-C}$$

A. A B + C / B C * A C − / −
B. A + B C / B C * A C − / −
C. A B C + / B C * A C − / −
D. A B + C / B * C A C − / −
E. A B + C / B C * A − C / −

**3. 比特位操作**

计算下列表达式：

10100 OR 01100 AND 01011

A. 01000
B. 11100
C. 00111
D. 11111
E. 00000

**4. 比特位操作**

计算下列表达式：

(LCIRC-2 (RSHIFT-1 (NOT 01110)))

A. 00010
B. 10001
C. 00001
D. 00100
E. 01100

**5. 此程序是怎样运行的？（循环）**

**执行下方**给定的输入值，以下程序的输出**值**是**多少**？

p = 1000, r = 10, t = 3

```
input p, r, t
o = p
for c = 1 to t
     i = p * r / 100
     p = p + i
next c
f = p - o
output f
```

A. 100
B. 110
C. 121
D. 331
E. 1331